

Solution Architecture

Streamlining IT procurement through standard laptop order automation

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Team ID	SWTID-2026-7609
Project Name	Streamlining IT procurement through standard laptop order automation
Maximum Marks	4 Marks

Purpose and recommended approach

This solution standardizes how employees request approved laptops and automates the checks, approvals, purchase handoff, and status updates around each order. A single catalog and policy-driven workflow reduce manual re-entry and make each decision traceable while leaving purchasing authority with the designated approver.

Business problem

Laptop requests commonly require repeated clarification of device specifications, eligibility, cost centre, approval, and delivery status. Email and spreadsheet handoffs can create delays, inconsistent orders, duplicate entry, and weak visibility. The target is a guided request that accepts only standard configurations and routes exceptions to a human reviewer.

Scope

- In scope: standard laptop catalog, requester and manager identity, eligibility and budget checks, approval routing, purchase-order handoff, supplier fulfilment status, notifications, and asset/endpoint registration.
- Out of scope: bespoke hardware sourcing, automated financial approval overrides, supplier selection policy, and device imaging or support procedures beyond the handoff.

Key design principle

Automate predictable standard orders; stop and route incomplete, out-of-policy, or exceptional requests for human review. No order is released to purchasing until required validation and approval are recorded.

Architecture and request flow

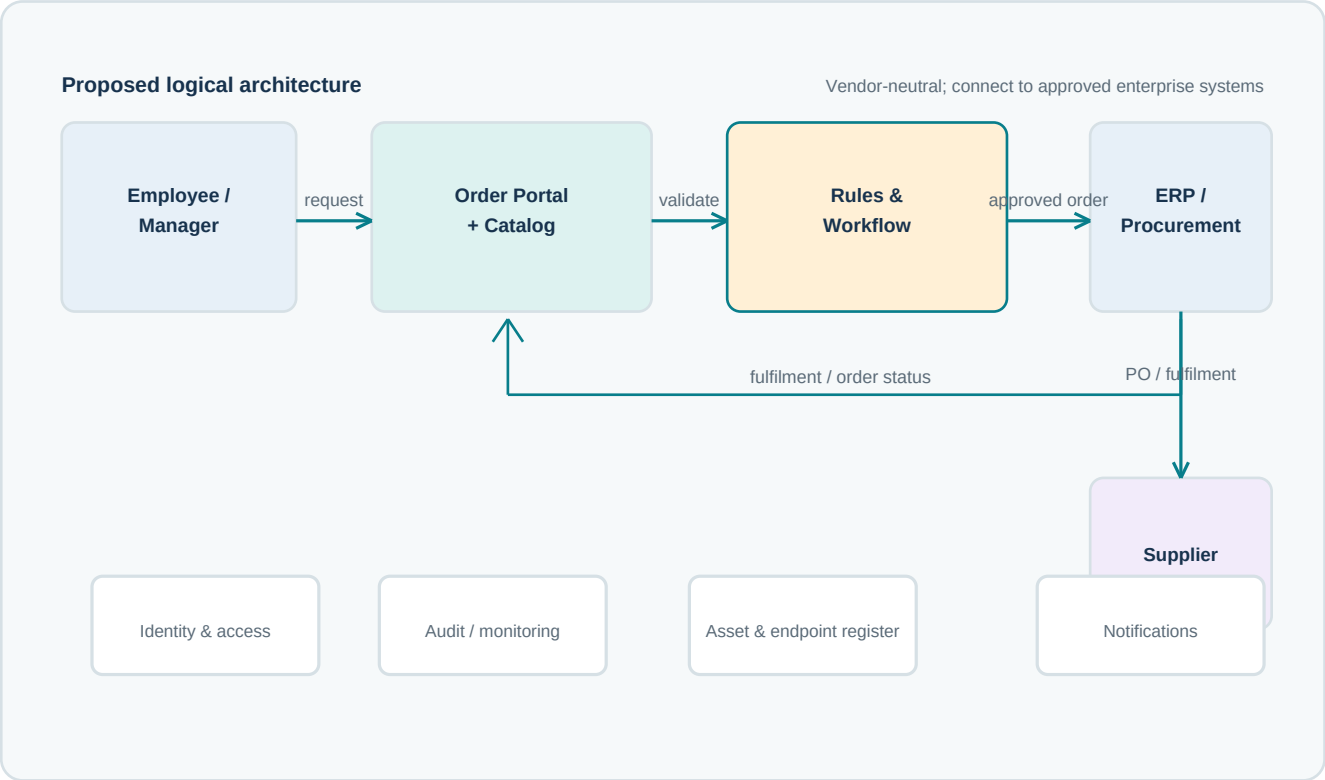


Figure 1. Proposed laptop order request, approval, procurement, and status flow.

End-to-end flow

Step	Action
1. Request	Employee signs in, selects an approved laptop configuration, provides business purpose and delivery location, and submits the request.
2. Validate	Workflow checks required fields, requester eligibility, catalog status, cost centre, and applicable budget/policy rules.
3. Approve	Manager or delegated approver reviews cost and policy context. Exceptions are held for procurement or IT review; rejected requests close with a reason.
4. Purchase	Approved request is sent to the procurement/ERP system as a purchase requisition or order handoff. The receiving system returns an identifier and processing outcome.
5. Fulfil	Supplier fulfilment updates are associated with the request. The employee receives status notifications; delivery triggers asset registration and the endpoint-management handoff.
6. Close	The request is closed when delivery/asset handoff is confirmed, or when cancellation/rejection is recorded. Audit events remain linked to the request ID.

Core components

Component	Responsibility
Request portal and catalog	Guided request form; displays only approved models, options, standard accessories, and policy guidance.
Workflow and rules	Validates fields and policy, routes approvals, enforces state transitions, and prevents duplicate or premature purchasing handoff.
Procurement / ERP connector	Maps approved requests to the enterprise purchasing interface; handles acknowledgements, retries, and reconciliation.
Supplier and fulfilment interface	Receives purchase and delivery state through the approved supplier/ERP integration or controlled manual update.
Asset and endpoint systems	Receives device and ownership details after fulfilment so inventory and device-enrolment processes can continue.
Notification and audit services	Sends requester/approver updates and records time-stamped decisions, integration outcomes, and state changes.

Controls and quality requirements

Security and data handling

- Use enterprise single sign-on and role-based access. Requesters see their own requests; approvers see assigned decisions; procurement and IT users receive only the access needed for their duties.
- Protect data in transit and at rest using enterprise-approved controls. Store only business and delivery information needed to process the order; restrict access to delivery details.
- Record who submitted, changed, approved, rejected, or transmitted each request, with timestamps and correlation IDs. Do not place secrets or unnecessary personal data in logs.
- Validate and authorize every integration call. Use managed credentials, encrypted connections, least-privilege service accounts, and monitored credential rotation.

Non-functional requirements

Area	Requirement / acceptance measure
Reliability	A request must remain recoverable through retries and reconciliation. A failed integration must not silently duplicate a purchase order; idempotency keys or equivalent duplicate protection are required.
Availability	The portal and workflow should follow enterprise availability and maintenance targets. Target values are to be agreed with IT before implementation.
Performance	Catalog browsing and submission should feel responsive under expected business-hour load. Establish numeric response-time and concurrency targets during detailed design.
Auditability	Every state transition and approval has an actor, timestamp, outcome, and linked request identifier; records follow the organization retention policy.
Usability	Forms explain required fields and validation errors; requesters can see current status and next action without contacting support.
Maintainability	Catalog, routing rules, approver mappings, and policy thresholds are configurable and have an owned change process.

Key risks and mitigations

Risk	Mitigation
Catalog or policy is stale	Assign catalog and policy owners; version changes and review them on a set cadence.
ERP/supplier interface is unavailable	Queue approved handoffs, retry safely, alert operations, and provide reconciliation before release or resubmission.
Approval routing is incorrect	Use maintained role/manager data; test routing scenarios and provide a controlled exception queue.
Fulfilment data is incomplete	Define required status fields and ownership; permit audited manual updates when an integration is unavailable.

Delivery plan and acceptance

Implementation phases

Phase	Deliverable
1. Discover and confirm	Confirm catalog, approval policy, ERP interface, supplier status source, asset/endpoint handoff, data classification, owners, and numeric service targets.
2. Configure and integrate	Build the guided request, workflow states, policy checks, approval routing, integrations, notifications, access roles, audit events, and operational dashboards.
3. Pilot	Run a limited group and standard laptop range. Test approval, rejection, exceptions, duplicate submissions, integration failure/retry, cancellation, fulfillment, and asset handoff.
4. Roll out and operate	Expand in stages after acceptance; train requesters and approvers; monitor queue age, failure/retry rates, exception volumes, and order completion. Assign support ownership and change control.

Acceptance criteria

- A requester can submit a complete standard laptop order from the approved catalog and receive a unique request ID.
- Ineligible, incomplete, out-of-policy, and exception requests are stopped or routed for review; no unapproved request reaches purchasing.
- An authorized approver can approve or reject with a recorded outcome and reason; the requester receives the result.
- An approved order is handed off once, and the procurement acknowledgement is linked to the request.
- Status changes, failures, retries, and manual interventions are traceable; duplicate protection and reconciliation are demonstrated.
- Fulfilment can be closed with an asset/endpoint handoff, and users can see the current status.

Decisions to confirm before build

The organization should confirm its system of record for the catalog, approval thresholds and delegates, procurement/ERP integration method, supplier status source, endpoint and asset interfaces, retention period, availability and response-time targets, and operational support owner. These items are intentionally left as decisions because the source material did not specify them.

Expected outcome

A controlled, auditable path for standard laptop orders that reduces manual coordination and gives employees and approvers a consistent view from request through delivery.